

MATT SCHUMACHER

New York, NY | (248) 990-3548 | ummatt@umich.edu

LinkedIn: [linkedin.com/in/ummatt](https://www.linkedin.com/in/ummatt) | Personal Site: mattschumacher.dev | GitHub: github.com/mattschu22

EDUCATION

University of Michigan, Ann Arbor, MI Graduated May 2026
M.S.E. in Computer Science, College of Engineering GPA: 4.0/4.0
Select Courses: 582 Adv. Operating Systems; 583 Adv. Compilers; 585 Adv. Scalable Systems for AI

University of Michigan, Ann Arbor, MI Graduated May 2025
B.S.E. in Computer Science, College of Engineering, Summa Cum Laude GPA: 3.9/4.0
B.B.A. in Business Administration, The Ross Business School, Magna Cum Laude
University Honors and Dean's List – All Semesters; Awarded William J. Branstrom Prize
Select Courses: 482 Operating Systems; 491 Distributed Systems; 489 Networking; 445 Machine Learning; 498 Machine Learning Research; 498 LLMs; 484 Databases; 476 Data Mining; 485 Web Systems; 493 UX Design

PROJECTS

Anthropic Performance Challenge January 2026
Produced solution that ran in 1184 cycles, placed 50th globally, and surpassed the best-known Opus 4.5 auto-research solution by 179 cycles, a 13% improvement. Challenge consisted of optimizing SIMD/VLIW assembly code.

Agentic Kernel Development Assistant August 2025 – April 2026
Designed and implemented a Linux patch generation agent that utilizes reproducer files from the Syzkaller kernel fuzzing project to autonomously test, generate, and apply fixes to the Linux kernel through a production GCP pipeline.
Achieved a 2x accuracy improvement over the baseline through a custom harness with compilation and reproducer feedback and a custom agentic architecture with dedicated tooling.

Database-Centric Operating System May 2026 - Current
Built an end-to-end SQL execution engine with parsing, binding, planning, query optimization, execution, persistence, and crash recovery, with focus on database/OS co-design for improved scheduling, memory management, and I/O performance with reduced overhead.
Instrumented SQL engine with performance metrics and dashboards using Perfetto, Grafana, and Linux perf tools to track performance improvements/regressions, identify bottlenecks, and guide optimization work.

KV-Cache Affinity Router January 2026 - Current
Reduced multi-cluster LLM inference prefill time by 82% through load-aware, KV-cache-aware routing with compact Bloom-filter-based cluster-state transmission for low-latency distributed routing.

EXPERIENCE

Center for Healthcare Engineering and Patient Safety || *Software Engineering Intern* May 2024 – August 2025
Reduced C++ backend simulation runtime by 78% through profiling, parallel execution, cache-aware optimization, and optimized numerical kernels; later led delivery of a TypeScript simulation tool with an optimized event loop now used across 170+ VA hospitals.

Aptiv || *Software Engineering Intern* May 2023 – July 2023
Architected and deployed a greenfield ML-driven UX analytics service for kiosk applications, spanning client-side telemetry collection, gRPC networking, backend data ingestion, and automated reporting.

Aptiv || *Data Science Intern* May 2022 – August 2022
Developed and implemented a multi-modal machine learning system achieving 91% accuracy in autonomous drive test failure detection, substantially streamlining testing procedures.

Lakeside Software || *Software Engineer* June 2018 – August 2021
Developed data mining strategies that linked user experience degradation symptoms with causative patterns, resulting in one of the first proactive IT technologies. Contributed to Best of Show Award at VMworld 2018 for Digital Workplace.

PATENTS AND PAPERS

Granted [US Patent No. 11,461,212](#), "Apparatus and Method for Determining the Underlying Cause of User Experience Degradation," and continuation-in-part patent [No. 11,983,088](#)
Co-author for "[Developing a systems-focused tool for modeling lung cancer screening resource needs](#)", describing my technical contributions at the Center for Healthcare Engineering. PubMed ID: 39237997

SKILLS AND ABILITIES

Computer Languages: C, C++, Go, Python, JavaScript, TypeScript, SQL
Backend/Infra: AWS, GCP, Docker, Linux, REST APIs, CI/CD, Perfetto, PostgreSQL, MongoDB, FastAPI, RabbitMQ
Proficiency in: System Design, Infrastructure, Distributed Systems, Synchronization, Concurrency, Networking, Databases